
CSI132 - Discrete Structures II
Tutorial 12 - Number Theory

1. Evaluate $155 \pmod{19}$ and $-221 \pmod{23}$
2. Find $a \pmod{m}$ and $a \operatorname{div} m$ when
 - i. $a = 9009, m = 223$
 - ii. $a = -10101, m = 333$ and $9009 \operatorname{div} 223 = 40$
3. Suppose that a and b are integers, $a \equiv 4 \pmod{13}$, and $b \equiv 9 \pmod{13}$. Find the integers c with $0 \leq c \leq 12$ such that:
 - i. $c \equiv 9a \pmod{13}$
 - ii. $c \equiv a + b \pmod{13}$
4. Prove or disprove. The product of any two even integers is a multiple of 4.
5. Prove or disprove. For all integers a, b and c , if $a \mid b$ and $a \mid c$, then $a \mid (2b - 3c)$.
6. Prove or disprove. If $a \mid b$ and $b \mid a$, where a and b are integers, then $a = b$.
7. Prove or disprove. If $a \mid bc$, where a, b and c are positive integers and $a \neq 0$, then $a \mid b$ or $a \mid c$.
8. Show that if n is an integer, then $n^2 \equiv 0 \pmod{4}$ or $n^2 \equiv 1 \pmod{4}$.